

GLOBAL MARKETS ANALYST

Updating Our G10 Term Premium Estimates — Still High

- We introduce new term premium estimates for G10 economies that utilize surveyed data on short rate expectations. These complement our existing estimates that are based purely on the yield curve.
- The new estimates that incorporate survey information typically show a lower and more stable term premium than those derived purely from market data. While both have value, we think the survey-based estimates likely better reflect the recent structural shift in rate expectations, and thus current levels of term premium in major bond markets.
- Japan is a clear illustration of the advantages of survey-based models. Yield curve-only estimates interpret almost the entirety of the yield curve steepening as a rise in term premium, whereas the inclusion of surveys better identifies the structural rise in policy rate expectations.
- Yield curve-only term premium estimates tend to be more volatile and better at gauging changes in real time. However, they also tend to be better predictors of future excess returns to owning duration over the medium term, which suggests that yield curve-only measures offer useful signals on future returns.
- We think understanding the variation and drivers of term premium is more important than the precise level for market participants. All our estimates show a rise in term premium in recent years, partially due to the combination of macro risks and bond supply.

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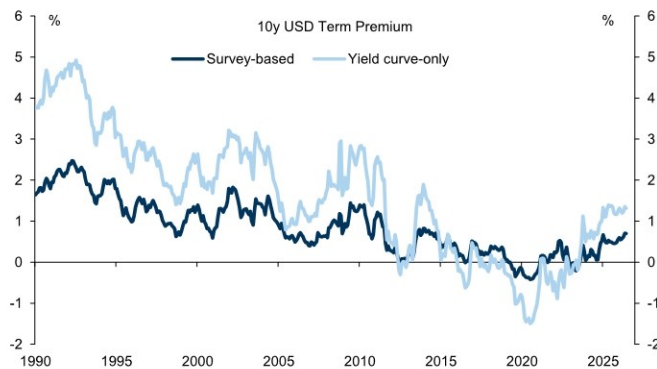
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Old Question, New Models

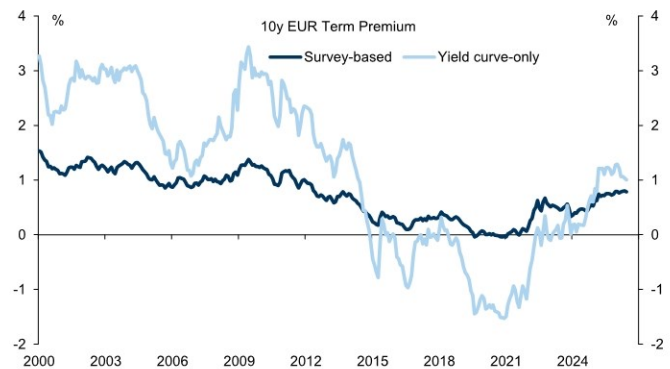
Rising global bond term premium has been a key feature of rates markets in recent years. To improve our analysis of term premium and its drivers, we introduce additional estimates to complement our current model. Our existing estimates are based purely on yield curve data, similar to the Adrian, Crump and Moench (ACM) model. Our new survey-based estimates are based on the Kim-Wright (KW) method, which uses survey information on short rate expectations to complement yield curve data. We roll out these estimates for each G10 curve, and will in future provide both survey-based and yield curve-only versions of the term premium estimates for each market. As we discuss below, we think that both models have value for market views, but that the new survey-based measures get a better grip on any structural shifts in rate expectations and as a result offer more economically intuitive measures for several markets.

Exhibit 1: Survey-based estimations currently sit below yield curve only term premium estimates ...



Source: Goldman Sachs Global Investment Research, Consensus Economics, Haver Analytics, Federal Reserve Board

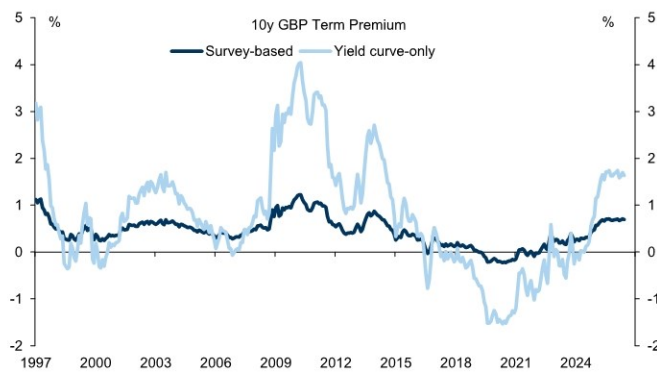
Exhibit 2: ... and also show relatively more stability



Source: Goldman Sachs Global Investment Research, Consensus Economics, Haver Analytics, Bloomberg

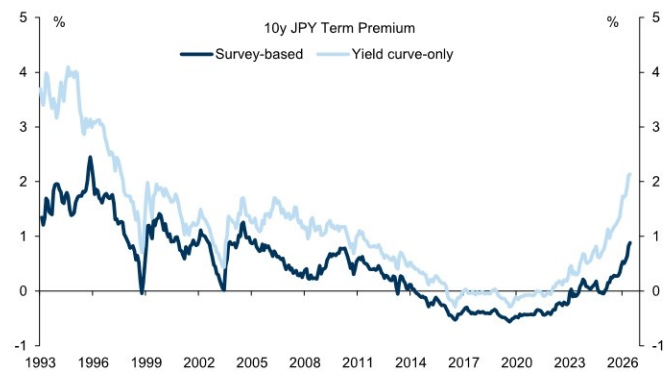
The approach of these models is qualitatively similar. They use information from the current yield curve – and surveys in the case of the KW method – to project forward estimates of the short rate path. This then allows us to build a spot curve based on these projections, usually described as the risk neutral rate expectations path. Essentially, the difference between the observed yield curve and this path is the term premium estimate of the model. Our new estimates are shown in ([Exhibit 1](#), [Exhibit 2](#), [Exhibit 3](#), and [Exhibit 4](#)) for G4, and [Exhibit 5](#) for the smaller G10 markets, with full details on method, input data and estimates in the Appendix.

Exhibit 3: UK survey-based estimates are the lowest among the G4, indicating an important part comes from high rates expectations



Source: Goldman Sachs Global Investment Research, Consensus Economics, Haver Analytics, Bloomberg

Exhibit 4: Using surveys shows that the rise in JGBs is not only a term premium story

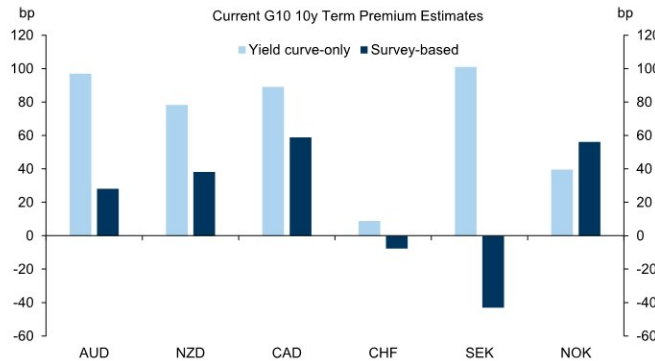


Source: Goldman Sachs Global Investment Research, Consensus Economics, Haver Analytics, Bloomberg

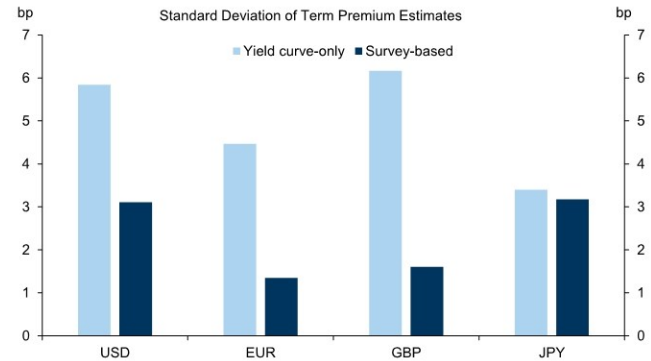
For most curves, our survey-based term premium estimates are lower and less volatile than the yield curve-only model. In the US, the survey-based approach estimates 10y term premium is 70bp, 40bp lower than the yield curve-only estimates. Similarly, in Europe the survey-based 10y term premium is around 15bp lower than yield curve-only estimates. The UK, which for most part of the last year had the highest term premium among the G4 (but has since been overtaken by Japan) by the yield curve-only estimates instead is showing the lowest level of term premium among peers on the survey-based estimates. The lower survey-based term premium therefore implies a higher rate expectation path for each market.

Great Survey Expectations

In the yield-curve only estimates, expectations are inferred primarily from the information embedded in the yield curve itself. In the survey-based specification, survey data directly discipline the projections for the short rate. This matters because when surveys point to a lower-for-longer or a structural rise in the policy path, the survey-based model will attribute more of the shift in the level of long-dated yields to expected future short rates and less to term premium. The yield curve-only based estimates rely more heavily on the long-run mean as an anchor for rate expectations, making them considerably more sensitive to the estimation window. That results in less volatility in the survey-based term premium estimates versus the yield curve-only estimates ([Exhibit 6](#)).

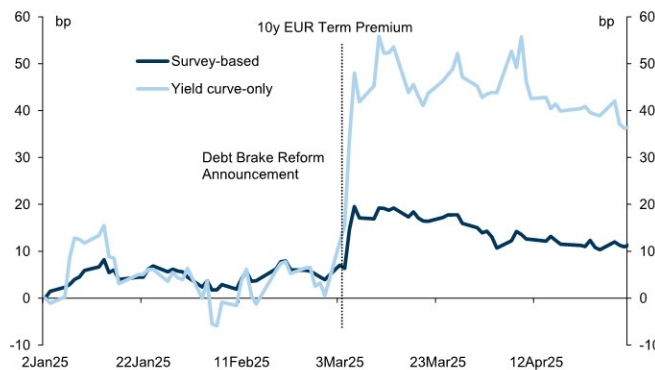
Exhibit 5: Within some of the smaller G10 the level of term premium differs meaningfully depending on approach

Source: Goldman Sachs Global Investment Research, Consensus Economics, Haver Analytics, Bloomberg

Exhibit 6: Our existing estimates of term premium are more volatile than survey-augmented estimates

Source: Goldman Sachs Global Investment Research

A recent illustration was the aftermath of the German debt brake reform in March 2025; the survey-based estimates reported a lower increase in term premium than the yield curve-only, as markets adjusted not just to higher bond supply, but also incorporated an upgraded growth outlook that justified a structurally higher path for real rates ([Exhibit 7](#)).

Exhibit 7: The debt brake removal might be a justification for higher real rates over time

Source: Goldman Sachs Global Investment Research

Exhibit 8: Including surveys suggests a larger role for rates expectations in the recent increase in JGB yields
Change in 10y JP rates expectations

Source: Goldman Sachs Global Investment Research, Consensus Economics

Another example of this is in Japan. Since 2023, surveyed policy rate projections have been rising alongside the convergence of longer-term inflation expectations towards 2%. Market pricing of OIS swap rates have moved to similar or even higher levels on a two-to-five year horizon. At the same time, the JGB yield curve has steepened significantly alongside a shrinking BoJ balance sheet and the market repricing duration risk in Japan. The yield curve-only model attributes bear steepening to a rise in term premium—short rates are moving only gradually, and based on the prior history (especially in Japan) these models will usually not show a rise in future short rates without some clearer sign of rises in the near-term, too.

In contrast, our survey-based model shows a more substantial rise in rate expectations – and correspondingly a smaller rise in JGB term premium – given the surveyed expectations have risen substantially. ([Exhibit 8](#)).

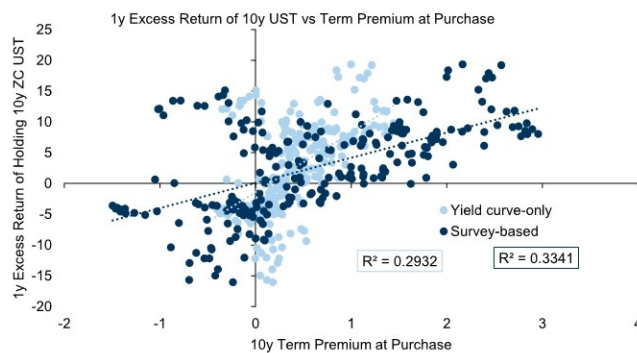
Survey-Based Estimates Capture Structural Shifts, but Yield Curve-Only Models Still Useful

The inclusion of surveys to estimate the expected short rate path is particularly helpful amid structural shifts in rate expectations that are less well picked up by yield curve models. We think that makes survey-augmented models more economically intuitive when judging the level of market expectations about future rates. At the same time, survey-based models can be slower to adjust to day-to-day changes in expectations given the lower frequency (usually monthly) of surveys – in the debt brake reform example above, that pickup in rate expectations became only visible once surveys for the month of March became available, with a notable delay.

We also find that the constraint imposed by the inclusion of surveys means that yield curve-only estimates tend to do a better job of predicting future excess returns to holding duration over the medium term ([Exhibit 9](#) and [Exhibit 10](#)). This suggests that purely market-based term premium measures offer greater value for trading signals.

Exhibit 9: Term premium is typically a useful predictor of duration excess returns over medium horizons

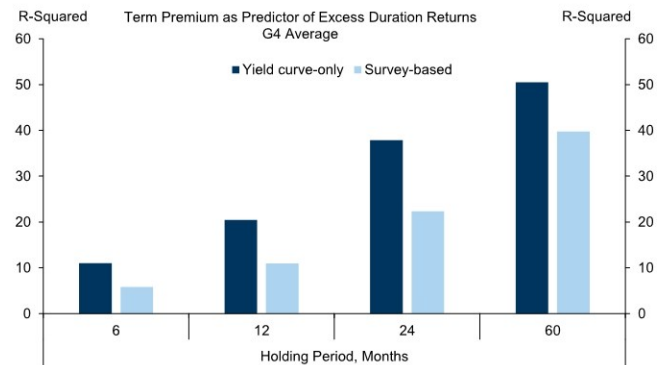
1y holding excess return of 10y UST vs term premium



Source: Goldman Sachs Global Investment Research, Goldman Sachs FICC and Equities

Exhibit 10: Yield curve-based (GS) approach better at predicting excess duration returns than survey-augmented estimates (KW)

G4 Average

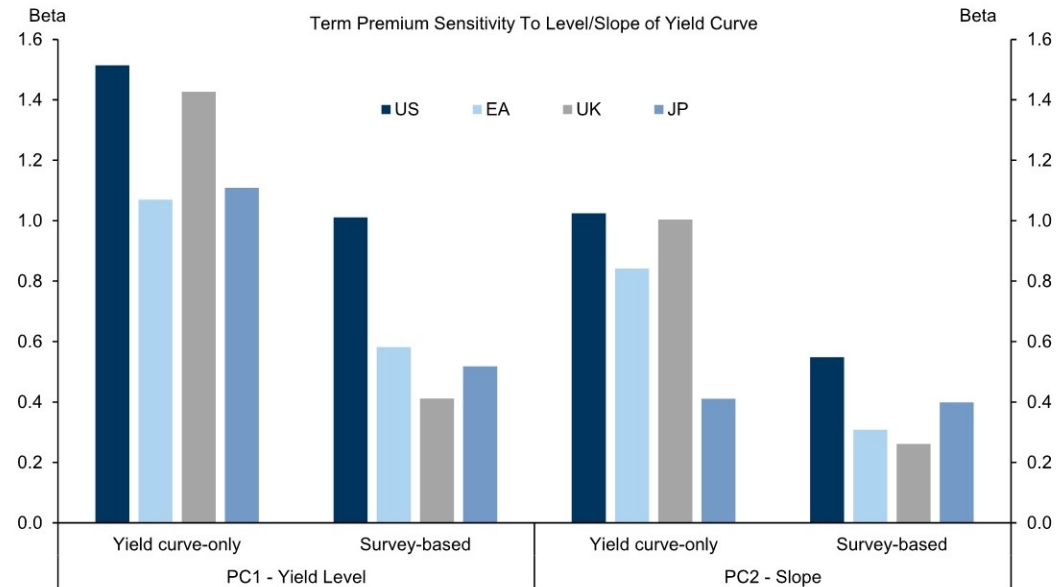


Source: Goldman Sachs Global Investment Research, Goldman Sachs FICC and Equities

The different approaches also show different responses to yield changes: for any given shock to the level of rates, the yield curve-only approach will result in a greater response in term premium than the survey-based estimates. The same holds true for a given steepening of the curve ([Exhibit 11](#)). Shifts in the curvature have no notable difference in impact on the different models. That helps to assess the *relative* behavior during rate shocks.

Exhibit 11: Yield curve models only show more responsiveness to shifts in level and slope

Term premium sensitivity (in pp) to a 1-standard deviation change in PC1 and 2



Source: Goldman Sachs Global Investment Research

Term Premium Still High Regardless of Estimate

Although there are key level differences between the survey-based and yield curve-only term premium estimates, we would not emphasise the precise level of either model's output. In our view, the value of these term premium estimates is to isolate drivers of movements in interest rates, and better understand the drivers of term premium. Currently, almost all models in all curves show that term premia have risen and sit near local highs, even as the precise levels vary. This implies only modest rises in rate expectations in most markets. The exceptions are Australia, the UK and Japan, which have seen sharper increases in rate expectations especially in the survey-based estimates.

However, the rise in G10 term premium in recent years does reflect changes in these key macro drivers over the same period, namely cyclical risk, the rise in inflation risk, the rise in the level and volatility of interest rates, high bond supply and a reversal of central bank balance sheet expansion ([Exhibit 12](#) and [Exhibit 13](#)). While elevated bond supply may still be contributing to higher term premium, recent swap spread widening points to macro explanations around growth, inflation and fiscal risks rather than just bond-specific weakness. In recent years we have found that term premium has remained elevated despite a decline in rates volatility – this high premium, low volatility combination points to a structural rise in duration risk premium beyond (counter)cyclical fluctuations, which typically affect term premium variation. In upcoming research we will dive deeper into the drivers and valuation of G10 term premium, using and expanding our existing suite of valuation frameworks.

Exhibit 12: G4 term premium has risen alongside shrinking central bank balance sheets

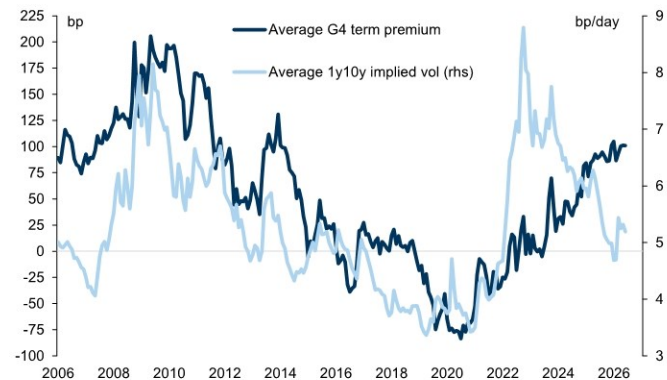
Average of yield curve-only and survey-based term premium across G4 (GDP weighted) and average central bank holdings (GDP weighted)



Source: Goldman Sachs Global Investment Research, Haver Analytics

Exhibit 13: The recent decline in implied volatility has not yet been reflected in lower term premium

Average of yield curve-only and survey-based term premium across G4 (GDP weighted) and average implied vol (GDP weighted)



Source: Goldman Sachs Global Investment Research, Goldman Sachs FICC and Equities

Appendix

We complement our current yield curve-only term premium model with a Kim-Wright-style model with surveys. Both models sit within the Gaussian affine class but differ on how the expectations component of long-dated yields is identified.

JSZ & ACM yield curve-only models

Our existing model is a no-arbitrage affine term structure model estimated from the cross-section of zero-coupon yields, based on the Joslin, Singleton and Zhu (JSZ) method. A set of three latent yield-curve factors is used to fit the curve and to model the evolution of expected short rates, allowing long-dated yields to be decomposed into an expectations component and a residual term premium.

Rather than estimating by maximum likelihood from the cross-section of yields as per our baseline model, the ACM model identifies the price of duration risk directly from realised excess bond returns. As these returns are generated from the yield curve itself, the information set remains the same, and so despite the different economic presentation, the results between our baseline and ACM models are very similar.

Kim-Wright survey-based model (KW)

The Kim-Wright-style model is also an affine term structure model but adds survey information to help identify the expected path of short rates. Survey forecasts provide additional, less noisy information about the expected path of short rates and help discipline this part of the model.

The model uses a Gaussian affine term structure model with three unobserved latent factors X_t which follow a VAR (1) under the physical measure:

$$X_{t+1} = \mu + \Phi X_t + \Sigma \varepsilon_{t+1}$$

The corresponding risk-neutral dynamics are governed by (μ^Q, Φ^Q) and the short rate is

affine in the factors like the ACM and the baseline model:

$$r_t = \delta_0 + \delta_1' X_t$$

Model implied zero-coupon yields are obtained from the standard affine no-arbitrage recursion, where the loadings are functions of the risk-neutral dynamics, the shock covariance matrix and the short rate parameters ($\mu^Q, \Phi^Q, \Sigma, \delta_0, \delta_1$). Relative to the baseline, where the only observation equations link the latent factors to the observed yield curve, Kim–Wright augments the observation vector with survey forecasts of the short rate.

$$\begin{pmatrix} y_t^{(n)} \\ s_t^{(h)} \end{pmatrix} = \begin{pmatrix} \text{Observed yields}_t \\ \text{Surveys}_t \end{pmatrix} = \begin{pmatrix} A_n + B_n' X_t + \eta_t^{(n)} \\ E_t^P[r_{t+h}] + \eta_t^{(h)} \end{pmatrix}$$

The survey row comes from treating a survey forecast of the 3-month rate h-months ahead as a noisy observation of the model-implied expected short rate at that horizon:

$$E_t^P[r_{t+h}] = \delta_0 + \delta_1' E_t^P[X_{t+h}]$$

Because the state follows a VAR (1) under the physical measure, the expected future state is:

$$E_t^P[X_{t+h}] = \Phi^h X_t + \sum_{j=0}^{h-1} \Phi^j \mu$$

Combining these two expressions gives the survey-observation equation:

$$s_t^{(h)} = \delta_0 + \delta_1' \left(\Phi^h X_t + \sum_{j=0}^{h-1} \Phi^j \mu \right) + \eta_t^{(h)}$$

The yield rows are affine pricing equations, while the survey rows are noisy observations of expected future short rates under the physical dynamics. For the long-horizon survey, the model maps the expected average 3-month rate over the relevant window into the survey observation. In practice, this anchors the persistent component of expected short rates while allowing for survey measurement error.

In our G10 implementation, the yield panel uses fitted monthly zero-coupon yields from 3M to 10Y. Survey inputs are Consensus Economics forecasts of the 3-month rate at short horizons (3m ahead and 1y ahead), plus the quarterly 6–10y average of long-horizon 3-month-rate forecasts where available. Surveys are included only when observed.

Estimation is carried out by maximum likelihood with a Kalman filter on the joint observation vector of yields and available survey forecasts. Once the parameters are estimated, the decomposition is the same as in the baseline model. The fitted yield is generated from the risk-neutral affine recursion, while the expectations component is the average expected future short rate under the physical dynamics:

$$TP_t^{(n)} = y_t^{(n)} - \frac{1}{n} \sum_{j=0}^{n-1} E_t^P[r_{t+j}]$$

Daily estimates are obtained by extracting daily factors from daily yields using the

estimated affine loadings and then applying the estimated model parameters to those daily factors.

Global Interest Rate Strategy

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